

Applied A.I. Solutions

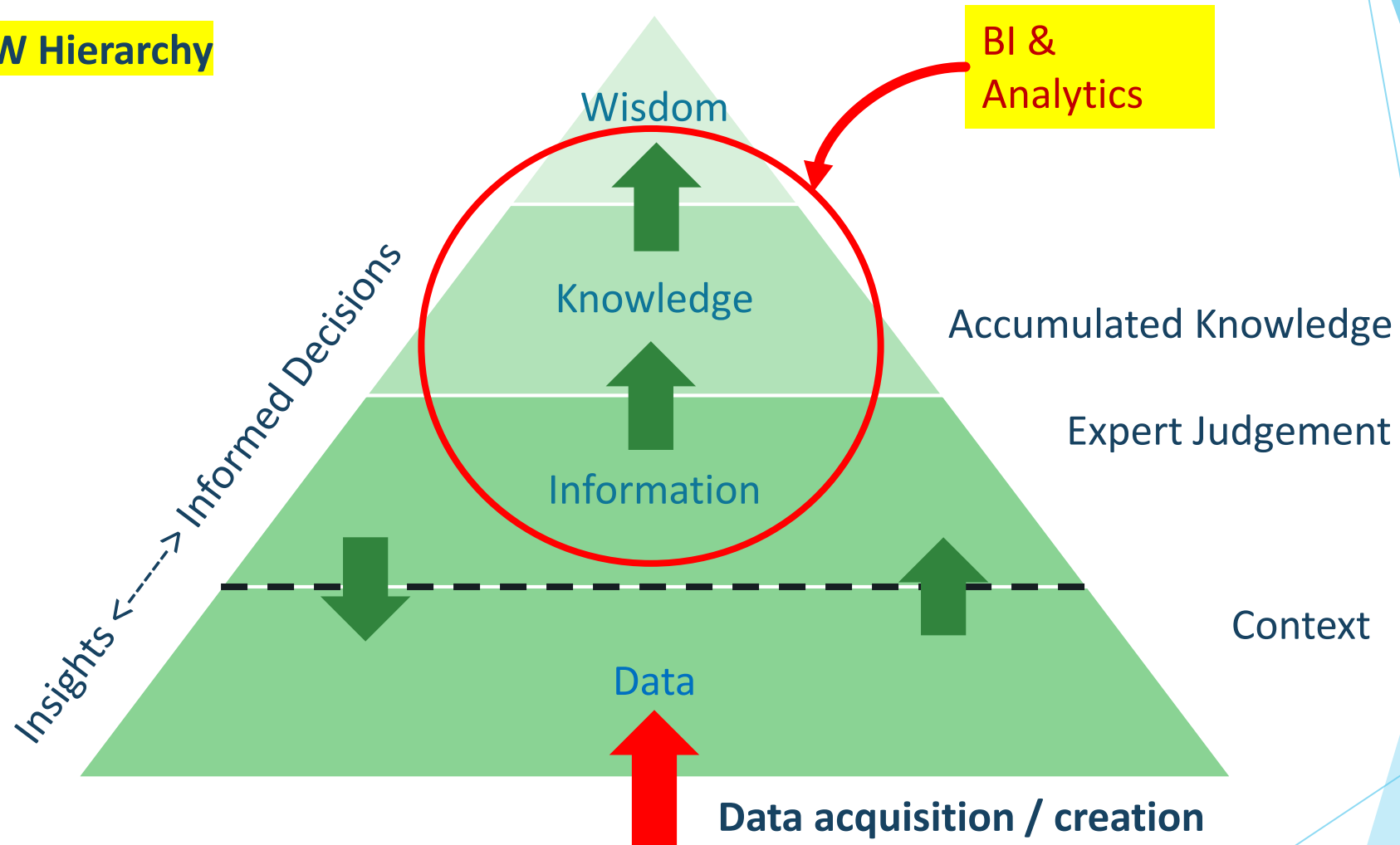
Data Visualization Techniques

Professor
Daniel Vitaver, EMBA

daniel.vitaver@georgebrown.ca

Data Value Extraction, Process Automation, Collaboration, Communication and Research

The DIKW Hierarchy



Data Visualization Techniques

Data Visualization Techniques and Tools help to
to understand data

to empower analytics

- to uncover hidden patterns and trends
- to see the story that lives within that data

to get relevant insights for ...

**Data
Value
Extraction**

- informed decision-making
- performance management
- innovation, research and development
- monitoring and control
- risk management
- business development
- enterprise asset valuation

Essential Concepts

- The **value** of data
- The importance of **context** and **storytelling**
- Introduction to **data types** (structured and unstructured)
- **Process Automation**, Collaboration, Communication, Research and Decision Making

Data types and main characteristics

A data type is an **attribute of data** which tells the compiler or interpreter how the programmer intends to use the data.

Common data types:

- Integer numbers (whole numbers)
- Floating-point or real numbers (contains decimal points)
- Characters (single letters)
- Strings (combination of letters, characters and symbols)
- Booleans (True/False, yes/no)

defines ...

- set of values
- constrains
- operations and business rules

Data types	
Uninterpreted	Bit, Byte, Word, Bit array
Numeric	Complex, Decimal, Fixed point, Floating point, Double precision, Extended precision, Long double, Integer, signedness, Interval, Rational
Pointer	Address, physical, virtual, Reference
Text	Character, String
Composite	Algebraic data type, generalized, Array, Class, Dependent, Equality, Inductive, Intersection, List, Object, metaobject, Option type, Product, Record, Set, Union
Logical	Boolean, Collection, Enumerated type, Exception, Function type, Recursive data type, Semaphore, Stream, Type class, Unit type
Object-oriented	Abstract data type, Data structure, Generic, Kind, metaclass, Object type, Parametric polymorphism, Primitive data type, Protocol, interface, Subtyping, Type constructor, Type conversion, Type system, Type theory, Variable

DBMS Classification (design or use)

Data Type

Structured

Semi-structured

Unstructured

DB Type

Relational (SQL)

Object-oriented

NoSQL

Model

Relational

Object-oriented

**Star schema /
Multidimensional**

Content Type

Real-time

Distributed

Data Warehouse

Processing Type

OLTP

OLAP

Batch

Real-time

Parallel

In-memory

Data Types

- **Structured Data**

- Structured data refers to all types of data that are **organized** in a any defined form
- Structured data is **defined types of data** in a structure
- Structured data lives in **rows and columns** and it can be **mapped into pre-defined fields**.
- Structured data has a **pre-defined data model**, typically a **relational** data model (SQL database)

Data Types – cont'd

- **Unstructured Data**
 - Data that either **does not have a pre-defined data model** or is **not organized in any pre-defined format**. Usually, it is stored in its native format.
 - It contains text, videos, audio, images, dates, numbers, other facts.
 - Common Techniques & Methodologies for **structuring text** are:
 - Natural Language Processing (NLP)
 - Text Analytics
 - Semantic Tagging with metadata

Data Types – cont'd

- **Semi-structured Data**

- Mix between structured and unstructured data
- Data has some consistent characteristics but doesn't conform to a rigid structure expected with a relational database
- Some organizational properties such as semantic tags or metadata is used to make it easier to organize

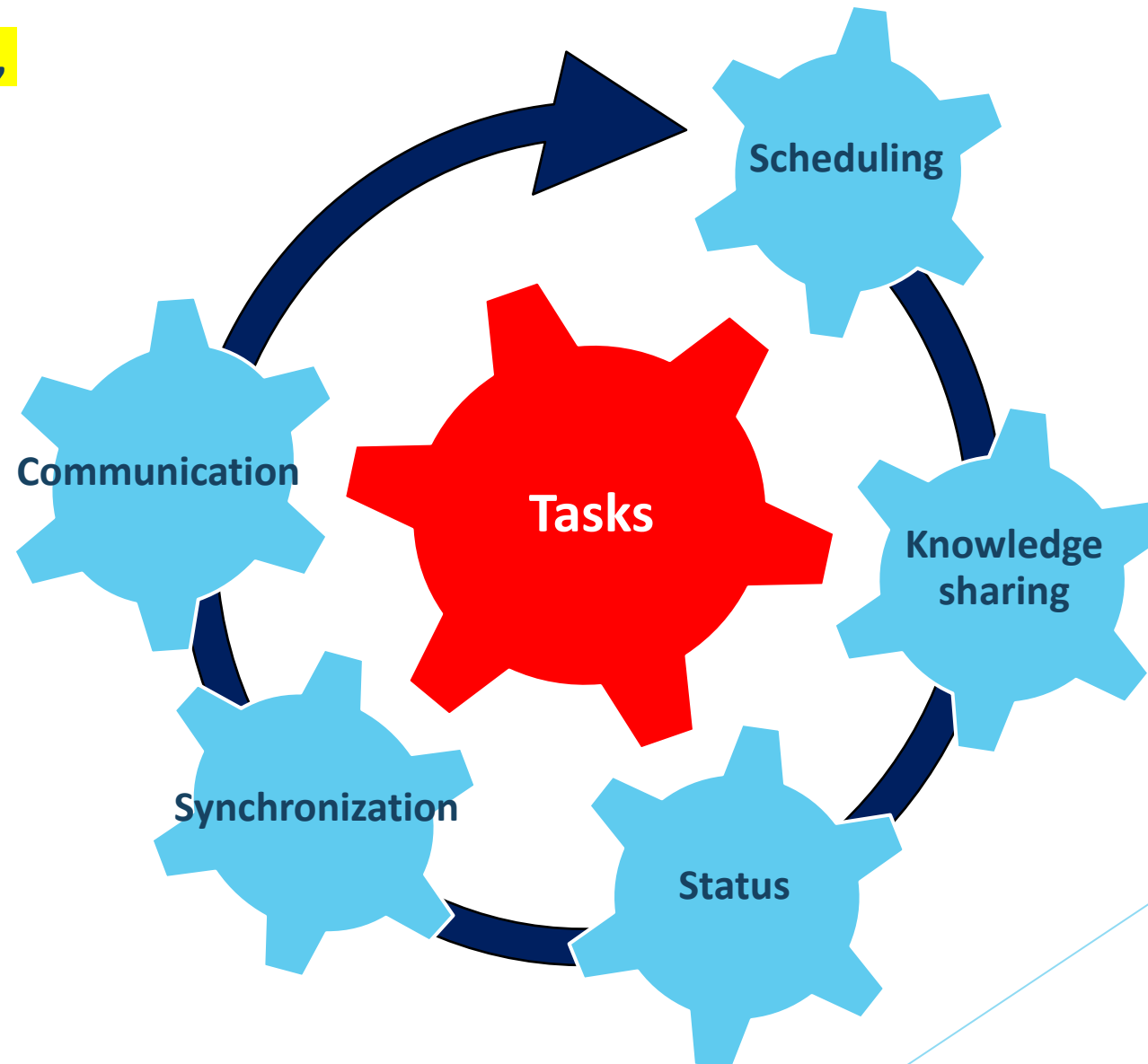
Business Processes

- A process is a set of interrelated **actions** and **activities** performed to achieve a pre-specified product, result, or service
- Each process is characterized by its **input**, the **tools and techniques** that can be applied, and the resulting **outputs**
- Each process can be broken down into other processes, and so on, until we reach a **level** in which simple activities are defined using measurable inputs and measurable outputs

Process Automation,

Collaboration,

Communication



Components



Process Automation Systems

- A process automation system (**PAS**) **automatically control a processes**
- A network is used to interconnect sensors, controllers, operator terminals
- Software and controls regulate equipment to run efficiently, ensuring the consistency of quality, and forecast when maintenance is needed

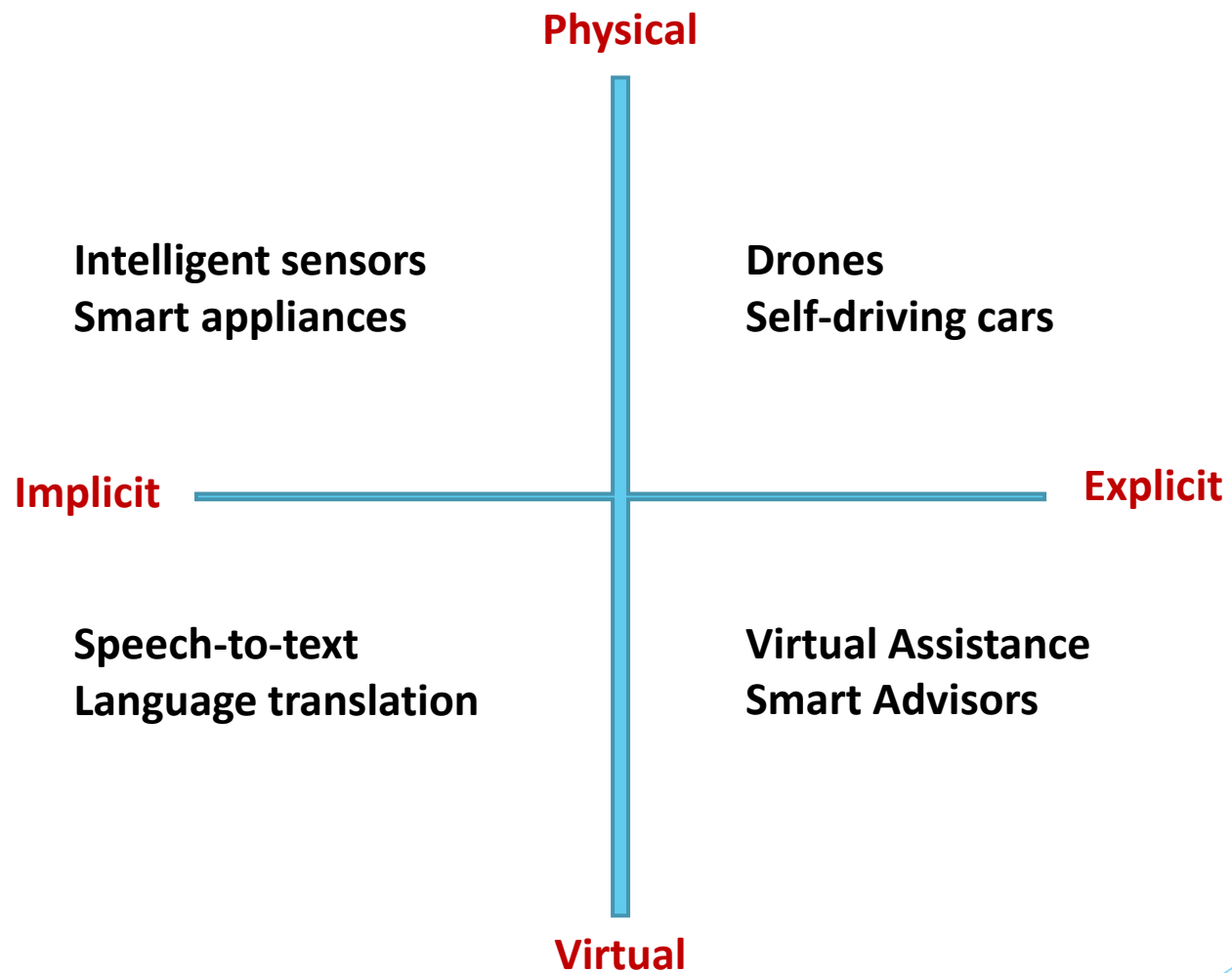
IT Process Automation (ITPA)

- ITPA **orchestrates** and integrates tools, people, processes through automated workflows
- ITPA uses software applications that are programmed to **perform any repeatable pattern, task or business workflow**

Digital Process Automation (DPA)

- Plain old automation
- Natural language processing
- Machine learning,
- **DPA is not AI/ML, but** fosters the use of AI/ML to make it more powerful
- Repetitive tasks done on a predefined cadence or sequence
- Tasks span multiple tools or cross-tool integrations

Applying AI to process automation



Process Automation - Current State

- Process automation is critical for the success of digital transformation. It's a key component of the IT strategies
- Complex Processes incorporate many steps and components across different technologies that need be automated
- Companies are looking to increase investment in process automation
- **Robotics** Process Automation provide relief as a noninvasive form of integration

Reference

- Read Camunda's report on "The State of Process Automation 2023"
- Read Gartner's report on "Move Beyond RPA to Deliver Hyperautomation, Dec 2021."

Collaboration and Communication – main features

- Unified communication channels
- Organized updated information
- Improved information accessibility
- Shared content
- Improved task management system
- Improved teams' collaboration with workgroups
- Deployed Team Collaboration / Workflow application software

Research and Decision-making

4-type research utilization (Nutley *et al* 2003)

1. Instrumental use - research feeds directly into decision-making
2. Conceptual use - research changes professionals' understanding providing new ways of thinking
3. Mobilization of support - research as an instrument of persuasion
4. Wider influence - research findings may come into use through networks of professionals and researchers and alter paradigms, strategies and actions

Information for decision-making - Risks

- Bias
- Recording and editing
- Selection (data vs information, lack of metadata)
- Timing
- Analysis and interpretation (selection of inappropriate techniques)

Annex